PRECOMPOSITION ON CONTINUOUS ALTERNATING MAPS IS NOT ANALYTIC IN CHARACTERISTIC 2

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ABSTRACT. For normed spaces E, E', F over a nontrivially normed field k and a finite set ι, precomposition $f \mapsto (m \mapsto m \circ (f, \dots, f))$ sends continuous linear maps $E \to E'$ to continuous linear maps between spaces of continuous alternating maps. It is C^n for every $n \in \mathbb{N} \cup \{\infty\}$ in every characteristic [1], and analytic when $|\iota|!$ is invertible in k (by polarisation). We show that it is *not* analytic in general: over every complete ultrametric field of characteristic 2 that is not spherically complete, with $E = E' = \ell^\infty$ and $|\iota| = 2$, analyticity fails at 0. The proof is a finite parity identity for determinants in characteristic 2, glued along ℓ^∞; the failure is witnessed by two coordinate projections of ℓ^∞. The result is formalized in Lean 4/Mathlib in the form "the general C^ω statement implies `False`".

1. STATEMENT

Notation. For normed spaces, $\mathcal{L}(E, E')$ is the space of continuous linear maps with the operator norm, and $\mathrm{Alt}^\iota(E, F)$ is the normed space of continuous alternating ι-linear maps $E^\iota \to F$, i.e. those vanishing whenever two arguments coincide; we write Alt^2 when $|\iota| = 2$. For a polynomial f and a monomial μ, $\mathrm{coeff}_\mu f$ is the coefficient of μ in f. We write $c_0 \subset \ell^\infty$ for the null sequences, $\mathbf{1}_S$ for the indicator of a set S, and $\mathbf{1} = \mathbf{1}_{\mathbb{N}}$.

Throughout, k is a field which is complete for a nontrivial absolute value $|\cdot|$ satisfying the *ultrametric inequality* $|x + y| \le \max(|x|, |y|)$; we call such a field *ultrametric*. (Absolute values with this property are also called non-archimedean; this is a condition on the absolute value and is unrelated to non-Archimedean *ordered* fields.) The sup norms used below inherit the inequality. We assume that

(K1) $\mathrm{char}\, k = 2$;

(K2) k is not spherically complete: there is a sequence of closed balls $B(c_n, r_n)$ such that $|c_{n+1} - c_n| \le r_n$, $r_0 > r_1 > \cdots$ and $\bigcap_n B(c_n, r_n) = \emptyset$.

The balls in (K2) are balls in k, and they are nested by the ultrametric inequality: for $a \in B(c_{n+1}, r_{n+1})$ we have $|a - c_n| \le \max(|a - c_{n+1}|, |c_{n+1} - c_n|) \le \max(r_{n+1}, r_n) = r_n$. In characteristic 2 that inequality is in fact automatic: an absolute value satisfies it as soon as it is bounded on the multiples of 1, and here these are only 0 and 1. Such fields exist; an explicit one is given in §6. Let $E = \ell^\infty = \ell^\infty(\mathbb{N}, k)$ with the sup norm, a Banach algebra under pointwise multiplication; write $m_a \in \mathcal{L}(E, E)$ for multiplication by $a \in E$, so that $\|m_a\| = \|a\|$ (evaluate at $\mathbf{1}$), and e_i for the unit vectors. For $x, y \in E$ let

$$W(x, y) := x \otimes y + y \otimes x \in \ell^\infty(\mathbb{N} \times \mathbb{N}, k), \qquad W(x, y)_{pq} = x_p y_q + x_q y_p,$$

and let $F \subseteq \ell^\infty(\mathbb{N} \times \mathbb{N}, k)$ be the closed linear span of all $W(x, y)$. Every element of F is a symmetric matrix with zero diagonal (as $W(x, y)_{pp} = 2x_p y_p = 0$) and a uniform limit of

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matrices of finite rank (as $W(x,y)$ has rank ≤ 2). By the ultrametric inequality $\|W(x,y)\| \le \|x\|\|y\|$, and $W(x,x) = 2\,x\otimes x = 0$; so $W\colon E\times E\to F$ is a bounded alternating bilinear map of norm ≤ 1. Let

$$Q\colon \mathcal{L}(E,E)\to \mathcal{L}\big(\mathrm{Alt}^2(E,F),\mathrm{Alt}^2(E,F)\big),\qquad Q(f)(m) = m\circ(f,f).$$

Theorem 1. *Q is not analytic at 0. In particular Q is not C^ω.*

Theorem 2. *The statement "for all nontrivially normed fields k, all normed spaces E, E', F over k and all finite ι, precomposition $\mathcal{L}(E,E')\to\mathcal{L}(\mathrm{Alt}^\iota(E',F),\mathrm{Alt}^\iota(E,F))$ is C^ω" is false. So is the weaker statement in which k ranges only over complete ultrametric fields of characteristic 2, $E = E'$ and F range only over Banach spaces, and $|\iota| = 2$.*

Theorem 2 follows from Theorem 1 and the existence of a field with (K1) and (K2). The rest of this note proves Theorem 1 in four steps.

2. Step 1: analyticity forces a bilinear lift

Lemma 3. *Let $Q\colon X\to Z$ be a map between normed spaces with $Q(tf) = t^2Q(f)$. If Q is analytic at 0 then there is a bounded bilinear $P\colon X\times X\to Z$ with $P(f,f) = Q(f)$ for all f.*

Proof. Let $\sum_n p_n$ be a power series of Q at 0. For fixed f the one-variable power series $\sum_n t^n\,p_n(f,\dots,f)$ and the polynomial $t^2Q(f)$ agree for small t. Power series coefficients in one variable are unique over any nontrivially normed field, so $p_2(f,f) = Q(f)$. Take $P = p_2$. (No division by 2 occurs, and P need not be symmetric.) □

So assume that a bounded bilinear P with $P(f,f) = Q(f)$ exists, and put, for $a,b,x,y\in E$,

$$\Phi(a,b;x,y) := P(m_a,m_b)(W)(x,y)\in F.$$

Then Φ is 4-linear, and since $\|m_a\| = \|a\|$ and $\|W\|\le 1$ we have

$$\|\Phi(a,b;x,y)\|\le C\|a\|\,\|b\|\,\|x\|\,\|y\|,\qquad C := \max(1,\|P\|).$$

Moreover

$$\Phi(a,b;x,x) = 0,\qquad \Phi(a,a;x,y) = W(ax,ay). \tag{1}$$

Since $2 = 0$, the first identity gives the symmetry $\Phi(a,b;x,y) = \Phi(a,b;y,x)$. We will contradict the existence of such a Φ.

3. Step 2: a finite parity identity

Fix a finite block $I\subset\mathbb{N}$ of even size $2m$ and $a',b'\in E$ with $\|a'\|,\|b'\|\le 1$ (in Step 4 they will carry the earlier blocks). Let X_i, Y_i $(i\in I)$ be indeterminates. Expanding Φ multilinearly in $x = \sum_{i\in I} X_ie_i$ and $y = \sum_{i\in I} Y_ie_i$ defines polynomials

$$G_{pq} := \Phi(a'+x,\,b'+y;\,a'+x,\,b'+y)_{pq}\in k[X,Y]\qquad (p,q\in I),$$

whose values at points $x,y\in k^I$ (vectors supported on I) are the entries of an $I\times I$ matrix $G(x,y)$. The matrix G is symmetric with zero diagonal, each G_{pq} has degree ≤ 2 in X and ≤ 2 in Y, its part of bidegree $(2,2)$ is $g_{pq} := \Phi(x,y;x,y)_{pq}$, and $G_{pq}-g_{pq}$ has total degree ≤ 3.

Lemma 4. *For $p\ne q$ we have, in $k[X,Y]$,*

$$g_{pq}(Y,X) = g_{pq}(X,Y) + X_p^2Y_q^2 + X_q^2Y_p^2\qquad\text{and}\qquad g_{pq}(X,X) = 0.$$

Proof. A field with a nontrivial absolute value is infinite, so it suffices to check these at points $x, y \in k^I$. By the symmetry in the last two slots, $g(y,x) = \Phi(y,x;y,x) = \Phi(y,x;x,y)$, so by bilinearity in the first two slots and (1)

$$\begin{aligned} g(x,y) + g(y,x) &= \Phi(x+y, x+y; x, y) + \Phi(x,x;x,y) + \Phi(y,y;x,y) \\ &= W\big((x+y)x, (x+y)y\big) + W(x^2, xy) + W(xy, y^2). \end{aligned}$$

Expanding, $W((x+y)x, (x+y)y) = W(x^2,xy) + W(x^2,y^2) + W(xy,xy) + W(xy,y^2)$. The first and last terms cancel against the two other summands and $W(xy,xy) = 0$, which leaves $W(x^2,y^2)$, whose (p,q) entry is $x_p^2 y_q^2 + x_q^2 y_p^2$. Finally $g(x,x) = W(x^2,x^2) = 0$. □

Let τ be the automorphism of $k[X,Y]$ swapping $X_i \leftrightarrow Y_i$, and δ the substitution $Y_i := X_i$.

Lemma 5 (balanced coefficients)**.** *Let $f \in k[X,Y]$ with $\tau f = f$ and $\operatorname{char} k = 2$. For every $S \subseteq I$ the coefficient of $\prod_{i\in S} X_i Y_i$ in f equals the coefficient of $\prod_{i\in S} X_i^2$ in δf.*

Proof. The coefficient of $\prod_S X_i^2$ in δf is the sum of the coefficients in f of all monomials $X^\alpha Y^\beta$ with $\alpha + \beta = 2 \cdot \mathbf{1}_S$. The involution $(\alpha,\beta) \mapsto (\beta,\alpha)$ of this set of monomials preserves coefficients (as $\tau f = f$), and its only fixed point is $\alpha = \beta = \mathbf{1}_S$; all other terms cancel in pairs. □

Proposition 6. *Let M be a perfect matching of I and $\nu := \prod_{i\in I} X_i Y_i$. Then*

$$\operatorname{coeff}_\nu \prod_{\{p,q\}\in M} G_{pq} = 1.$$

Proof. The monomial ν has total degree $4m$, the maximum possible, so only the top parts contribute:

$$\operatorname{coeff}_\nu \prod_M G_{pq} = \operatorname{coeff}_\nu \prod_M g_{pq}.$$

Put $g'_{pq} := g_{pq} + X_p^2 Y_q^2$. By Lemma 4,

$$\tau g'_{pq} = g'_{pq} \qquad \text{and} \qquad \delta g'_{pq} = X_p^2 X_q^2.$$

A product containing a factor $X_p^2 Y_q^2$ has no squarefree monomial, so replacing every g_{pq} by g'_{pq} does not change the coefficient of ν. By Lemma 5, applied to the τ-invariant product $\prod_M g'_{pq}$, this coefficient is the coefficient of $\prod_{i\in I} X_i^2$ in

$$\delta \prod_M g'_{pq} = \prod_{\{p,q\}\in M} X_p^2 X_q^2 = \prod_{i\in I} X_i^2,$$

namely 1. □

Lemma 7. *Let A be a symmetric matrix with zero diagonal over a commutative ring of characteristic 2, of even size $2m$. Then $\det A = \sum_M \prod_{\{p,q\}\in M} A_{pq}^2$, the sum being over perfect matchings; and the number of perfect matchings, $(2m-1)!!$, is odd.*

Proof. In $\det A = \sum_\sigma \prod_i A_{i\sigma(i)}$ (all signs are 1), σ and σ^{-1} contribute equally by symmetry, so only involutions survive; those with a fixed point contribute 0; a fixed-point-free involution is a perfect matching and contributes $\prod A_{pq}^2$. □

Lemma 8 (hypercube sum)**.** *Let R be a commutative ring of characteristic 2 and let $f \in R[Z_1, \dots, Z_N]$ have total degree $\le N$. Then $\sum_{z\in\{0,1\}^N} f(z)$ is the coefficient of $Z_1 Z_2 \cdots Z_N$ in f.*

Proof. By linearity we may take f to be a monomial Z^α. At a point $z \in \{0,1\}^N$ it takes the value 1 if $z_i = 1$ for all i in the support of α, and 0 otherwise; so the sum is $2^{N-|\operatorname{supp}\alpha|}$, which vanishes in R unless every variable occurs in Z^α. The only monomial of degree $\le N$ in which all N variables occur is $Z_1 \cdots Z_N$. □

Corollary 9. *The values of* $\det G$ *at the* $0/1$ *points sum to* 1,

$$\sum_{x,y\in\{0,1\}^I} \det G(x,y) = 1,$$

and consequently there are $x, y \in \{0,1\}^I$ *with* $|\det G(x,y)| \ge 1$.

Proof. For a perfect matching M the product $\prod_{\{p,q\}\in M} G_{pq}$ has total degree $\le 4m$, which is the number of variables X_i, Y_i. By Lemma 8 and Proposition 6 its sum over the $0/1$ points is therefore $\operatorname{coeff}_\nu \prod_M G_{pq} = 1$. Squaring is additive in characteristic 2, so by Lemma 7

$$\sum_{x,y\in\{0,1\}^I} \det G(x,y) = \sum_M \Big(\sum_{x,y\in\{0,1\}^I} \prod_{\{p,q\}\in M} G_{pq}(x,y) \Big)^2 = (2m-1)!! \cdot 1 = 1,$$

because $(2m-1)!!$ is odd. If all the values $\det G(x,y)$ had absolute value < 1, the ultrametric inequality would give $|1| < 1$. □

The point is that the coefficient 1 does not depend on the unknown lift P: all of its freedom is carried by τ-invariant terms vanishing on the diagonal $Y = X$, which Lemma 5 cannot see. No single pair (x, y) works for every lift, but the 2^{4m} vertices of the cube cannot all fail at once.

4. Step 3: locality on ℓ^∞

For $x \in \ell^\infty$ write $x|_{\ge N}$ for x with its coordinates below N set to 0.

Proposition 10. *Under* (K2)*, a bounded linear functional* φ *on* ℓ^∞ *with* $\varphi(e_i) = 0$ *for all* i *is zero.*

Proof. φ kills finitely supported sequences, so $|\varphi(x)| \le \|\varphi\| \sup_{i\ge M} |x_i|$ for every M. Suppose $\varphi \ne 0$. For every $0 < s < \|\varphi\|$ there is w with $\varphi(w) = 1$ and $\|w\| \le 1/s$; fix one such vector w_0. Because $r_{n+1} < r_n$ we can choose $s_n \in \ell^\infty$ with $\varphi(s_n) = c_{n+2} - c_{n+1}$ and $\|s_n\| \le r_n/\|\varphi\|$: take $s_n = (c_{n+2} - c_{n+1})w$ with $\|w\| \le r_n/(r_{n+1}\|\varphi\|)$. Let $x_n := c_1 w_0 + \sum_{k<n} s_k$; then $\varphi(x_n) = c_{n+1}$ and $\|x_{n+d} - x_n\| \le r_n/\|\varphi\|$ by the ultrametric inequality. The diagonal sequence $y_i := x_i(i)$ is bounded, since $\|x_i\| \le \max(\|x_0\|, r_0/\|\varphi\|)$, and $|y_i - x_n(i)| \le r_n/\|\varphi\|$ for $i \ge n$. Hence $|\varphi(y) - c_{n+1}| = |\varphi(y - x_n)| \le r_n$, and since $|c_{n+1} - c_n| \le r_n$ the ultrametric inequality gives $|\varphi(y) - c_n| \le r_n$ for every n. So $\varphi(y) \in \bigcap_n B(c_n, r_n)$, a contradiction. □

This is the direction of van der Put's theorem [2] that we need, $(\ell^\infty/c_0)' = 0$; the index shift absorbs the usual $(1+\varepsilon)$ loss, and no Hahn–Banach argument is used.

Proposition 11 (multilinear Phillips property)**.** *Let* λ *be a bounded* d*-linear form on* ℓ^∞ *and* $\varepsilon > 0$*. There is* N *such that* $|\lambda(u_1, \dots, u_d)| \le \varepsilon$ *whenever* $\|u_j\| \le 1$ *for all* j *and* $u_j = u_j|_{\ge N}$ *for some* j*.*

Sketch (the complete proof is in the Lean development)*. The case* $d = 1$, $\lambda = \varphi$. (a) $\varphi(e_i) \to 0$. Otherwise $|\varphi(e_{i_k})| \ge \varepsilon$ along a subsequence. Let T be the bounded operator with $T(x)_{i_k} = x_k/\varphi(e_{i_k})$ and all other coordinates 0, and $\psi := \varphi \circ T$, so that $\psi(e_k) = 1$ for all k. With sh the right shift, $\psi \circ (1 - \mathrm{sh})$ kills every e_k, hence vanishes by Proposition 10; but $(1 - \mathrm{sh})\mathbf{1} = e_0$, so $0 = \psi(e_0) = 1$. (b) For fixed x, $\varphi(x|_{\ge N}) - \varphi(x|_{\ge N+1}) = x_N \varphi(e_N) \to 0$, so $\varphi(x|_{\ge N})$ converges;

the limit is a bounded linear functional of x killing every e_i, hence it is 0. (c) If the claim failed for ε, then (b) would give unit vectors w_k with disjoint finite supports moving to infinity and $|\varphi(w_k)| > \varepsilon$; the functional $c \mapsto \varphi(\sum_k c_k w_k)$ (coordinatewise sum) then contradicts (a).

The case $d > 1$, by induction on d. Suppose the claim fails for ε. First, for every M there is a unit tuple with $|\lambda| > \varepsilon$ *all* of whose slots vanish below M: start from a witness one of whose slots is supported very far out, and split any other slot into its part below M, a finite combination of the e_i, and its tail; the inductive hypothesis, applied to the finitely many $(d-1)$-linear forms obtained by freezing that slot at e_i, $i < M$, shows that the part below M contributes at most ε, so the tail carries $|\lambda| > \varepsilon$. Next, by (b) each slot can be cut down to a finite window. Finally choose such tuples $z^{(s)}$ with windows moving to infinity fast enough and let x be the coordinatewise sum of their first slots. Then $\lambda(x, z_2^{(s)}, \dots, z_d^{(s)})$ equals $\lambda(z^{(s)})$, plus finitely many terms with an earlier first slot (each $\le \varepsilon$ by the inductive hypothesis for $\lambda(z_1^{(r)}, \cdot)$, $r < s$), plus one term with a far first slot ($\le \varepsilon$ by the case $d = 1$). So it has absolute value $> \varepsilon$ for every s, contradicting the inductive hypothesis for the $(d-1)$-linear form $\lambda(x, \cdot)$. □

This is exactly where ℓ^∞ and (K2) are used; the statement is false for c_0 and for spherically complete fields.

5. Step 4: gluing

Choose inductively blocks $I_1 < I_2 < \cdots$ with $|I_t| = 2t$ and vectors $a_t, b_t \in \{0,1\}^{I_t}$ as follows. Given the earlier stages let $a' = \sum_{s<t} a_s$, $b' = \sum_{s<t} b_s$, and apply Corollary 9 to get a_t, b_t with $|\det G_t| \ge 1$, where $G_t = \Phi(a' + a_t, b' + b_t; a' + a_t, b' + b_t)|_{I_t \times I_t}$. Put $\varepsilon_t := \frac{1}{2} C^{-2t}$ and use Proposition 11 (with $d = 4$) for the finitely many entry forms $\Phi(\cdot)_{pq}$, $p, q \in I_t$, to place all later blocks so far out that these forms are at most ε_t on unit arguments one of which is supported there.

Let $a, b \in \ell^\infty$ be defined by $a|_{I_t} = a_t$, $b|_{I_t} = b_t$ and zero elsewhere. These are not norm-convergent sums; $a = \mathbf{1}_A$ and $b = \mathbf{1}_B$ are indicator functions of two subsets of $\mathbb{N}$, so m_a, m_b are coordinate projections. Let $\mathcal{O} := \Phi(a, b; a, b) \in F$. Splitting $a = a^{\le t} + a^{>t}$, $b = b^{\le t} + b^{>t}$ and using multilinearity, $\|\mathcal{O}|_{I_t \times I_t} - G_t\| \le \varepsilon_t$. All entries are bounded by C. Expanding the determinant multilinearly in columns, every term containing a column of the perturbation is at most $\varepsilon_t C^{2t-1} < 1$, so the ultrametric inequality gives

$$|\det \mathcal{O}|_{I_t \times I_t}| = |\det G_t| \ge 1 \qquad \text{for every } t.$$

On the other hand $\mathcal{O} \in F$ is within $\eta := (2C)^{-1}$ of a matrix R of some finite rank r. Expanding $\det(R + (\mathcal{O} - R))$ on $I_t \times I_t$ in the same way, terms with more than r columns from R vanish, so $|\det \mathcal{O}|_{I_t \times I_t}| \le C^r \eta^{2t-r} < 1$ once $t > r$. This contradiction proves Theorem 1. □

6. An explicit field

Let k_0 be any field of characteristic 2; the simplest choice is $k_0 = \mathbb{F}_2$, which is what the Lean development takes. For $j \ge 0$ let $k_0((t^{1/2^j}))$ be the field of *formal* Laurent series $\sum_{n \ge N} a_n t^{n/2^j}$ ($N \in \mathbb{Z}$, $a_n \in k_0$) in the single variable $t^{1/2^j}$: only finitely many exponents are negative, unlike for the Laurent series of complex analysis, and this is what makes the product of two series well defined. These fields form an increasing chain, and we let

$$\mathcal{P} := \bigcup_{j \ge 0} k_0((t^{1/2^j})), \qquad |f| := 2^{-\operatorname{ord} f},$$

be the field of 2-*power Puiseux series*. Here, for $f = \sum_q a_q t^q \ne 0$, $\operatorname{ord} f$ is the smallest exponent q with $a_q \ne 0$ (it exists because the exponents of f are bounded below and lie in

some $2^{-j}\mathbb{Z}$), and we put $\operatorname{ord} 0 = +\infty$, $|0| = 0$. The lowest terms of two series multiply, so $\operatorname{ord}(fg) = \operatorname{ord} f + \operatorname{ord} g$ and $|fg| = |f||g|$; and $f + g$ has no term below $\min(\operatorname{ord} f, \operatorname{ord} g)$, so $\operatorname{ord}(f + g) \geq \min(\operatorname{ord} f, \operatorname{ord} g)$, which is the ultrametric inequality $|f + g| \leq \max(|f|, |g|)$. The absolute value is nontrivial because $|t| = \frac{1}{2}$. Every element of $\mathcal{P}$ lies in one of the fields $k_0((t^{1/2^j}))$, so its exponents have a common denominator 2^j; the bound j depends on the element, exactly as for ordinary Puiseux series $\bigcup_n k_0((t^{1/n}))$. For $k_0 = \mathbb{F}_2$ (or any perfect k_0) the field $\mathcal{P}$ is the perfection of $k_0((t))$, because $k_0((t))^{1/2^j} = k_0((t^{1/2^j}))$. Let k be the completion of $\mathcal{P}$; the absolute value extends to it with the same properties, and k has characteristic 2, so (K1) holds.

For (K2) let $\gamma_m = 1 - 2^{-(m+1)}$, $c_n = \sum_{m \leq n} t^{\gamma_m} \in \mathcal{P}$ and $r_n = 2^{-\gamma_{n+1}}$: then $|c_{n+1} - c_n| = r_n$, the radii decrease strictly, and $r_n > \frac{1}{2}$. If some $a \in k$ lay in all the balls, pick $y \in \mathcal{P}$ with $|a - y| < \frac{1}{2}$; then y lies in all of them too. But $y \in k_0((t^{1/2^j}))$ for some j, so its exponents lie in $2^{-j}\mathbb{Z}$, while γ_j has denominator 2^{j+1}. So for $n \geq j$ the series $y - c_n$ has a nonzero coefficient at t^{γ_j} and $|y - c_n| \geq 2^{-\gamma_j} > r_n$, a contradiction.

Informally, the only candidate for a common point of the balls is the formal sum $\sum_m t^{\gamma_m}$. Its exponents have unbounded denominators and all lie below 1, so its terms do not tend to 0 and the partial sums c_n are not a Cauchy sequence; the sum exists in the (spherically complete) Hahn series field $k_0((t^{\mathbb{Q}}))$ but not in k. Completing $\mathcal{P}$ does add series with unbounded denominators, such as $\sum_m t^{m+2^{-m}}$, but only with exponents tending to ∞: below any bound, the exponents of an element of k still have a common denominator. (The completion of an algebraic closure of $\mathbb{F}_2((t))$ is another example of a field with (K1) and (K2).)

7. Remarks

Remark 12 (Why characteristic 2, why ℓ^∞, why this F). In characteristic $\neq 2$ the lift $\frac{1}{2}\big(m \circ (f, g) + m \circ (g, f)\big)$ exists, and the parity argument breaks (pairs add instead of cancelling). If E has an orthogonal base (e_i), e.g. $E = c_0$, and F is Banach, then

$$P(f, g)(m)(x, y) = \sum_{i<j} (x_i y_j - x_j y_i)\, m(f e_i, g e_j)$$

is a bounded lift in every characteristic, so Q is analytic; a similar formula works for a base of E', and t-orthogonal bases suffice. This covers finite-dimensional E and all Banach spaces over a discretely valued k [3], so the counterexample needs a dense valuation; one can also show that Q is analytic when F is spherically complete. On c_0 the glued vectors a, b of Step 4 do not exist and Proposition 11 fails. For $E = \ell^\infty$ and the larger target $\ell^\infty(\mathbb{N} \times \mathbb{N}, k) \supseteq F$ a bounded lift also exists (sort the indices entrywise), but it sends W to matrices such as the "staircase" $\mathbf{1}_{\{(p,q)\,:\,p \text{ odd}, q \text{ even}, p<q\}}$, which are not limits of finite-rank matrices, so it leaves F. These positive statements are not proved here and are not part of the formalization.

Remark 13 (What is forced). No single pair (x, y) works for every lift: adding to Φ a bounded 4-linear map with values in F that is alternating in (a, b) and in (x, y) gives another admissible Φ, and this can change a given off-diagonal entry of $G(x, y)$ by an arbitrary scalar. Blocks of bounded size cannot give a contradiction either: for $u = \mathbf{1}_{\text{odd}}$, $v = \mathbf{1}_{\text{even}}$ the rank-two matrix $W(u, v) \in F$ has the entry 1 in every diagonal block $\{2s - 1, 2s\}$. On a finite block the sorted lift of the previous remark has norm 1 and satisfies all the identities of Step 2, so there is no finite-dimensional obstruction: the invariant is the rank of one element of F against the size of the blocks, and Step 3 is what allows infinitely many blocks to be glued into one element.

Remark 14 (Higher degree, other characteristics). On paper, in characteristic 2 every $|\iota| \geq 2$ is covered by wedging with a coordinate of $E \oplus k$; this is not formalized. In odd characteristic ℓ, for forms of degree ℓ, the question is open: odd-degree forms have no Pfaffian-type top power, so a different invariant is needed.

8. Formalization

The Lean 4/Mathlib development is at

https://github.com/Deicyde/alternating-analytic-counterexample

The library `AltCounter/` contains no `sorry`, and the only axioms it uses are the three standard ones (`propext`, `Quot.sound` and `Classical.choice`). Its headline theorem assumes that Mathlib's

`ContinuousAlternatingMap.compContinuousLinearMapCLM`

is C^ω for all nontrivially normed fields, normed spaces and finite index types, and derives `False`; a second theorem does the same from the weaker statement in Theorem 2. Both are restated in a Mathlib-only file `Challenge.lean` (with `sorry` in place of the proofs, as that format requires) and checked against `Solution.lean` by the `leanprover/comparator` tool. Steps 1–4 and §6 are formalized, including the complete proof of Proposition 11 by induction on d; Lemma 8 is `sum_eval_cubePt`, and the field of §6 is built with $k_0 = \mathbb{F}_2$. The Lean constants in Step 4 are chosen with more slack than here, and the Lean form of Corollary 9 records only that the witnesses lie in the unit ball, which is all that Step 4 uses. Not formalized: non-analyticity at base points other than 0, index types with more than two elements, the positive statements in the remarks, and the bundle-level consequence (no C^ω vector-bundle structure on bundles of continuous alternating maps in this generality).

References

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